

DNA Replication Study Notes

Session Objectives:

SOM.MK.I.BPM1.1.FTM.3.BCHM.0035	Describe semi-conservative replication of DNA
SOM.MK.I.BPM1.1.FTM.3.BCHM.0036	Compare and contrast the similarities and differences between DNA Pol I and DNA Pol III of E. coli
SOM.MK.I.BPM1.1.FTM.3.BCHM.0037	Describe how DNA is synthesized from its 5' to 3' end
SOM.MK.I.BPM1.1.FTM.3.BCHM.0038	Explain the significance of the origin of replication during the process of DNA synthesis.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0039	Explain the problems associated with unwinding of the DNA double helix
SOM.MK.I.BPM1.1.FTM.3.BCHM.0040	Describe the role of helicase enzymes in unwinding DNA prior to its replication
SOM.MK.I.BPM1.1.FTM.3.BCHM.0041	Describe the role of single-strand binding protein & topoisomerases in DNA replication
SOM.MK.I.BPM1.1.FTM.3.BCHM.0042	Outline the order of events from RNA priming to completed DNA during replication in E.coli.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0043	Distinguish between synthesis of leading and lagging strands of DNA. Identify the significance of DNA ligase in lagging strand synthesis
SOM.MK.I.BPM1.1.FTM.3.BCHM.0044	Describe the significance of the various proteins and enzymes required for replication and predict the effects of mutations of these proteins and enzymes on DNA replication
SOM.MK.I.BPM1.1.FTM.3.BCHM.0045	Evaluate the proofreading activity of DNA polymerase
SOM.MK.I.BPM1.1.FTM.3.BCHM.0046	Differentiate between DNA replication in eukaryotes and prokaryotes
SOM.MK.I.BPM1.1.FTM.3.BCHM.0047	Outline the mechanisms of action of drugs that interfere with DNA replication

Assigned Reading: Lippincott's Illustrated Review Biochemistry, 8th Edition.

Chapter 30, [DNA Structure, Replication, and Repair](#) | [Lippincott® Illustrated Reviews: Biochemistry, 8e](#) | [Medical Education | Health Library](#) (lwwhealthlibrary.com)

- **Lippincott's Biochemistry 8th Edition**
<http://sgu.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/book.aspx?bookid=3073>

Questions 30.2 – 30.5. Concept Map on Page 479.

DNA Replication in Prokaryotes

With the structure of the DNA molecule being understood by the work of Watson and Crick, it became apparent that the molecules characteristics would facilitate its own replication during cell division.

The process of DNA synthesis is accomplished in both prokaryotes and eukaryotes by a group of enzymes called **DNA polymerase**.

The following are the minimum components necessary for DNA replication to occur:

1) All four dNTPS (deoxynucleoside triphosphates)

- these are the precursor molecules that are incorporated by DNA polymerase into the newly synthesized DNA strand
- energy for the synthesis reaction is provided by the cleavage of a nucleoside triphosphate to form a nucleoside monophosphate and pyrophosphate.
- each nucleoside contains a 5' phosphate group and a 3' hydroxy group (-OH) that contributes to the formation of the phosphodiester bond of the DNA backbone of the synthesized DNA.

2) A fragment of DNA

- the DNA polymerase reads a single DNA strand in the 3' to 5' direction that acts as a template for synthesis of the new DNA strand.

3) DNA polymerase

- all DNA polymerases synthesis DNA in the 5' to 3' direction while reading the template strand DNA in the 3' to 5' direction
- DNA polymerase synthesizes DNA by using the 3'-hydroxyl group (=OH) of the RNA primer as the acceptor of the first deoxyribonucleotide.
- the synthesis involves the generation of a phosphodiester bond between the 3'-OH group of the previously incorporated deoxynucleotide (or RNA primer) and the 5' phosphate group of the incoming deoxynucleotide.
- bases are incorporated into the growing DNA strand by using the template strand to select the appropriate complementary base to be incorporated into the synthesis

4) Magnesium Ions (Mg²⁺)

- magnesium ions are required as co-factors for DNA polymerase to be active

5) A primer providing a free 3'-OH group

-in order for DNA synthesis to be initiated an available 3'-OH group must be present and attached to the template DNA. This free 3'-OH group is either provided by an RNA primer in the case of initiation of replication, or by a previously incorporated deoxynucleotide.

Initiation of DNA replication

DNA replication starts first by the formation of a region of single stranded DNA resulting from the separation of the two strands of a double stranded DNA. This forms a region called the *replication fork* – the site where DNA replication and synthesis occurs. DNA replication in both prokaryotes and eukaryotes is initiated at sites on the chromosome called the *Origin of replication (ORI)*. These sites consist of short repeated DNA sequences that are A-T rich, and as a result the hydrogen bonding in this region of the chromosome is weak. Therefore, DNA strand separation can occur easily producing a region of single stranded DNA that can serve as a template for DNA synthesis. Prokaryote chromosomes possess a single origin of replication while eukaryote chromosomes may possess hundreds of ORI dispersed throughout the genome.

Steps for the Initiation of DNA Replication:

1) Binding of Initiator proteins (DnaA proteins) to the Origin of replication

-initiator proteins bind the ORI sequence causing strand separation and the formation of localized regions of single stranded DNA.

-initiator proteins only cause DNA strand separation during the initial step of DNA replication

2) Recruitment of DNA helicase (DnaB protein) to the replication fork

-DNA helicase binds to single stranded DNA at the replication fork and moves towards regions of double stranded DNA and unwinding the DNA to form a larger region of single stranded DNA. The energy required for this process is supplied by the hydrolysis of ATP to ADP.

-DNA helicase is responsible for all DNA strand separation after the initiation of DNA replication

3) Addition of an RNA primer to the template DNA by DNA primase

- DNA primase synthesizes a short RNA that is complementary to the DNA template.

-the RNA primer provides the free 3'-OH group for the synthesis of DNA

4) Synthesis of DNA from the primer by DNA polymerase (DNA polymerase activity)

-DNA polymerase initiates DNA synthesis by forming a phosphodiester bond between the 3'-OH group of the primer and an incoming deoxyribonucleotide and continues to synthesize DNA from incorporated nucleotides.

-DNA polymerase synthesizes DNA in a 5' to 3' direction

DNA synthesis – Leading and Lagging Strand:

The DNA polymerases that are responsible for synthesizing DNA are only able to “read” the parental template DNA sequence in the 3’ to 5’ direction while synthesizing the new DNA strands in a 5’ to 3’ direction. As a result, the two newly synthesized DNA strands are being synthesized in opposite directions with respect to the replication fork.

1. **Leading strand** – the strand that is being synthesized in the direction of the advancing replication fork is called the leading strand. The leading strand is synthesized continuously after the addition of the initial RNA primer.
2. **Lagging strand** – the strand that is being synthesized in the opposite direction or away from the replication fork is called the lagging strand. Synthesis of the lagging strand requires the continuous addition of an RNA primer as the replication fork advances. This produces a set of short DNA fragments being copied near the replication fork. These DNA fragments are called *Okazaki fragments*. The Okazaki fragments are approximately 1000 nucleotides long in prokaryotes and from 100-200 nucleotides long in eukaryotes. Later during the replication process these Okazaki fragments are joined to form a continuous DNA strand by the enzyme **DNA ligase**.

In prokaryotes, the enzyme DNA polymerase III accomplishes both the synthesis of the leading and lagging strand DNA.

Completion of DNA replication:

Following the completion of both leading and lagging strand DNA synthesis, the final step of the process involves the removal of the RNA primers on the lagging strand and the joining together of Okazaki fragments to produce a continuous strand of DNA.

The removal of the RNA primer and its replacement with DNA is carried out by the enzyme DNA polymerase I.

Once the RNA primer is removed and replaced with DNA, the enzyme DNA ligase generates a phosphodiester bond between adjacent Okazaki fragments. This produces a continuous DNA strand.

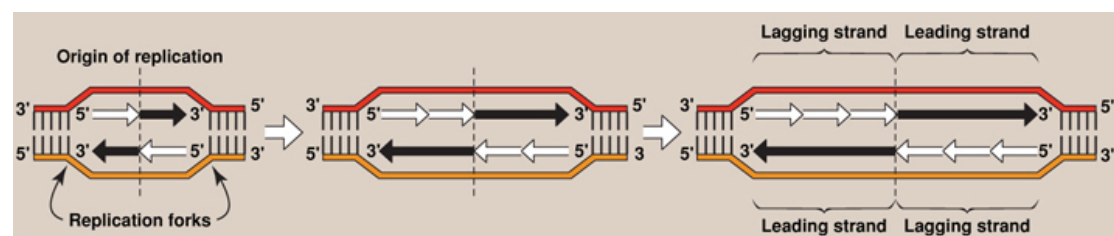


Figure depicting leading and lagging strand synthesis of DNA at a replication fork.

Additional players in DNA replication:

Single stranded binding protein

It is important that the separated single strands do not re-associate before they can be copied by the DNA polymerase. To prevent this happening, a special protein called single stranded binding protein (SSBP) attaches itself to the single stranded regions of the DNA after they have been separated. The SSBP molecules do not interfere with copying of the DNA template because they are displaced from the DNA by the movement of DNA polymerase.

Topoisomerases

When DNA unwinds in the cell, prior to replication, its ends are not free. In prokaryotes the ends are joined to each other to form a circle, and in eukaryotes, the linear DNA is fixed periodically to the chromosome scaffold to form many loops. Since the ends of the DNA are not free to rotate, as the DNA unwinds, positive supercoils are introduced into the DNA molecule. If these are not removed, they eventually exert so much force that the DNA strand separation stops. To prevent this, topoisomerases relieve the positive supercoiling. This occurs in both prokaryotes and eukaryotes. In prokaryotes, DNA gyrase is the topoisomerase that is involved.

Other Activities of DNA Polymerase:

In addition to the DNA polymerase activity that incorporates deoxynucleoside triphosphates into a growing DNA strand, DNA polymerase also has other important activities.

3' to 5' exonuclease activity

Both **DNA polymerase I and III** have 3' to 5' exonuclease activity. This activity has a proofreading (error checking) function. The enzyme removes nucleotides one at a time from the 3' end of the growing DNA chain. However, it will only remove a nucleotide if its base is not hydrogen bonded to the template. Failure of a base to form hydrogen bonding only occurs if the DNA polymerase makes a mistake and adds the wrong nucleotide to the end of the growing chain.

This error checking property greatly increases the fidelity of the DNA replication process. For example, DNA polymerase I will make one mistake in every 10^4 to 10^5 nucleotide incorporations. After 3' to 5' exonuclease error checking, the error rate is reduced to one mistake in every 10^7 incorporations. Other error checking mechanisms further reduce the overall error rate to 10^{10} incorporations.

5' to 3' exonuclease activity

DNA polymerase I also possesses 5' to 3' exonuclease activity. This activity removes nucleotides from the 5' end of the primer strand. It differs from the 3' to 5' exonuclease in that the substrate must be base paired. The exonuclease activity is greater if it is accompanied by simultaneous synthesis of DNA by the

polymerase. This 5' to 3' activity is responsible for removing the RNA primer that is added by DNA primase during DNA replication.

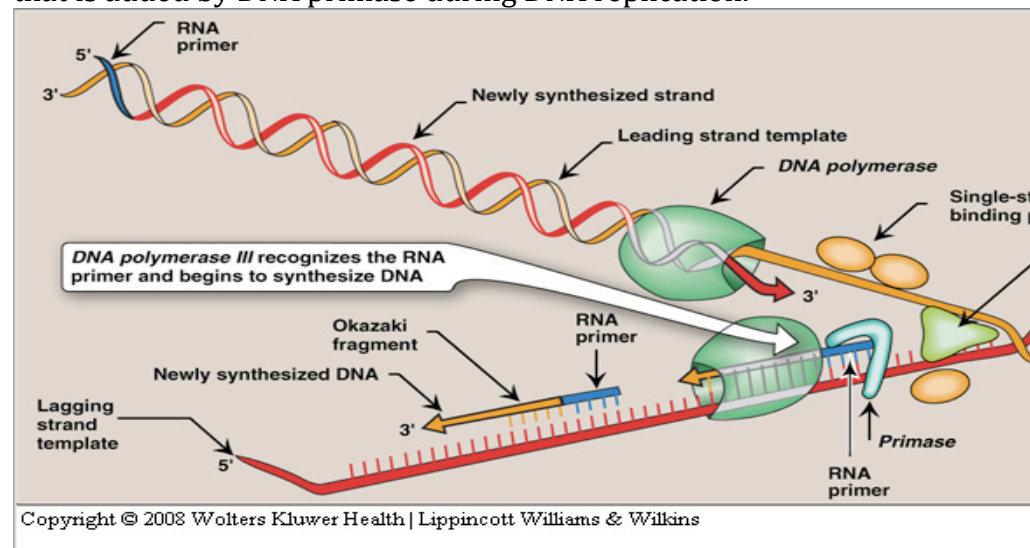


Figure showing the components of DNA replication at a replication fork.

DNA Replication in Eukaryotes

Much less is known about DNA replication in eukaryotes than in prokaryotes, but it is, in principle the same in eukaryotes, however the system is more sophisticated and involves more enzymes.

At least 5 DNA polymerases are found in eukaryote cells. In most cases these DNA polymerases have similar properties to the prokaryote enzymes.

E. coli DNA has 4,000,000 base pairs (bp). The smallest human chromosome has 46,000,000 (bp) and the largest one has 240,000,000 bp. If there were only one origin of DNA replication per chromosome it would take approximately 55 days to replicate the largest chromosome. Therefore, eukaryote chromosomes have multiple origins of replication per chromosome.

Telomeres and Telomerase

Cells of eukaryotes face a special problem of replicating the ends of their linear DNA molecules, compared to prokaryotes that have circular genomes. This is because, following the removal of the RNA primer from the 5' end of the lagging strand there will be a gap remaining in the DNA that can not be replicated.

To solve this problem, a special complex containing enzymatic proteins and an RNA molecule called telomerase operates at the ends of eukaryote chromosomes during replication.

The ends of each eukaryote chromosome called the telomeres, possesses several thousand non-coding repeated sequences – typically AGGGTT.

The complex, telomerase contains a protein that functions as a reverse transcriptase, as well as a short piece of RNA that can act as a template for DNA synthesis.

The RNA template found within the telomerase base pairs with the telomeric sequences in the 3' overhang at the ends of the eukaryote chromosomes. This base pairing occurs in an overlapping fashion such that only a portion of the RNA template is paired with the DNA. The reverse transcriptase then utilizes the RNA template to extend the telomeric sequence in the typical 5' to 3' direction and therefore further lengthening and extending the 3' overhang. The telomerase trans-locates again to the 3' overhang and this process is repeated many hundreds of times.

Once the 3' overhang of the DNA is lengthened significantly, primase can use it to synthesis an RNA primer and DNA polymerase can extend this primer and synthesize DNA in the "gap" as per normal DNA replication.

Clinical Relevance:

Actinomycin D is another antibiotic derived from *Streptomyces*. This antibiotic binds tightly to double stranded DNA and inhibits both DNA replication and transcription. It does this by preventing the formation of regions of single stranded DNA. It is effective in both prokaryotes and eukaryotes (will be mentioned during the transcription lecture).

Quinolone Family of drugs act by inhibiting DNA gyrase (bacteria topoisomerase II). As a result, this causes a halt in DNA replication due to the build up of positive supercoiling ahead of the replication fork during DNA replication.

Camptothecin inhibits the eukaryote enzyme topoisomerase I. The drug acts by preventing the religation of DNA following the cleavage activity of topoisomerase I. This ultimately leads to DNA damage and apoptosis. As a result this drug, and synthetic analogs have been used as chemotherapy to treat cancer.

Etoposide inhibits the eukaryote enzyme topoisomerase II. The drug acts by preventing the re-ligation of DNA following the cleavage activity of topoisomerase II. This ultimately leads to DNA damage and apoptosis. As a result, this drug, and synthetic analogs have been used as chemotherapy to treat cancer.

Interestingly, both camptothecin and etoposide compounds were first derived from tree and plant extracts – the Happy Tree (China) and Mayapple (North America), respectively. You may recall that "toxol", a mitotic inhibitor and anti-cancer agent, is also derived from a tree extract - the Yew tree (Europe).